

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285446

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Santillo; Vincent et al.

GUEST MEASUREMENT SYSTEMS AND METHODS

Abstract

A guest measurement system includes a controller associated with an attraction measurement area of an individual attraction within a theme park. The guest measurement system includes a controller configured to receive sensor signals indicative of guest characteristics from one or more sensors. The controller includes a processor configured to detect one or more limbs of a guest based in part on the sensor signals, identify a plurality of subregions for an individual limb of the one or more limbs, estimate dimensions for the identified plurality of subregions for the individual limb, and generate a guest height calculation based on the estimated dimensions for the identified plurality of subregions.

Inventors: Santillo; Vincent (Orlando, FL), Hirapara; Cole (Orlando, FL), Melo; Anthony (Orlando, FL), Henderson; Dionté Omar (Orlando, FL), Marceau; Taylor (Bryson City, NC), Gosbee; Katarina (Orlando, FL)

Applicant: Universal City Studios LLC (Universal City, CA)

Family ID: 83510840

Appl. No.: 19/216232

Filed: May 22, 2025

Related U.S. Application Data

parent US continuation 17715546 20220407 parent-grant-document US 12315262 child US 19216232

us-provisional-application US 63172442 20210408

Publication Classification

Int. Cl.: G06V20/52 (20220101); A63G31/00 (20060101); G01S17/08 (20060101); G01S17/88 (20060101); G06T7/60 (20170101); G06T7/73 (20170101); G06V10/74 (20220101);

U.S. Cl.:

CPC **G06V20/53** (20220101); **G01S17/08** (20130101); **G01S17/88** (20130101); **G06T7/60** (20130101); **G06T7/73** (20170101); **G06V10/761** (20220101); **G06V40/107** (20220101); **A63G31/00** (20130101); **G06T2207/30196** (20130101); **G06T2207/30232** (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/715,546 filed Apr. 7, 2022, entitled “GUEST MEASUREMENT SYSTEMS AND METHODS”, which claims priority to and the benefit of U.S. Provisional Application No. 63/172,442, filed Apr. 8, 2021, and entitled “GUEST MEASUREMENT SYSTEMS AND METHODS,” the disclosure of which is incorporated by reference in its entirety for all purposes.

BACKGROUND

[0002] The present disclosure relates generally to the field of amusement parks. More specifically, embodiments of the present disclosure relate to methods and equipment utilized to measure guests in an amusement park to assess one or more physical characteristics and/or determine eligibility to enter an attraction of the amusement park, e.g., enter an amusement ride as a passenger.

[0003] Theme or amusement parks have become increasingly popular. More sophisticated and creative ride attractions have been helpful in increasing the popularity and success of such parks. Certain attractions may be height-restricted to passengers that are above a height threshold. For example, theme park ride attractions may involve ride vehicles, such as roller coaster type cars, or other vehicles that move along a ride path. Height restrictions for a ride may involve restricting the ride to passengers within a particular height range that can be fitted to the restraints of each seat. The passengers can be measured before entering the ride to determine if they fit within the height restrictions. For example, a passenger can manually compare their height to a line on the wall.

SUMMARY

[0004] Certain embodiments commensurate in scope with the originally claimed subject matter are summarized below. These embodiments are not intended to limit the scope of the disclosure, but rather these embodiments are intended only to provide a brief summary of certain disclosed embodiments. Indeed, the present disclosure may encompass a variety of forms that may be similar to or different from the embodiments set forth below.

[0005] In accordance with one embodiment, a guest measurement system includes one or more sensors configured to generate sensor signals indicative of guest characteristics in an attraction measurement area of an individual attraction within a theme park. The guest measurement system includes a controller configured to receive the sensor signals indicative of the guest characteristics from the one or more sensors. The controller includes a processor configured to detect one or more limbs of a guest based in part on the sensor signals. The processor is configured to identify a plurality of subregions for an individual limb of the one or more limbs and to estimate dimensions for the identified plurality of subregions for the individual limb of the one or more limbs and generate a guest height calculation based on the estimated dimensions for the identified plurality of subregions.

[0006] In accordance with an embodiment, a method of measuring a guest in an attraction measurement area includes receiving one or more sensor signals indicative of a guest distance from a height marker from one or more sensors. In response to detecting that the guest is within a threshold distance from the height marker, a guest measurement is initiated. The guest

measurement includes using the one or more sensor signals to identify limbs of the guest, estimating dimensions of subregions of the identified limbs based on the height marker, and using the estimated dimensions of the subregions to generate a guest height calculation.

[0007] In accordance with one embodiment, a guest measurement system includes one or more sensors configured to generate sensor signals indicative of guest characteristics in an attraction measurement area of an individual attraction within a theme park, and a controller configured to receive the sensor signals indicative of the guest characteristics from the one or more sensors. The controller includes a processor configured to identify one or more anatomical features of a guest based in part on the sensor signals, estimate a dimension for the identified one or more anatomical features, and generate a guest height calculation based on the estimated dimension.

[0008] In accordance with one embodiment, a guest measurement device includes one or more sensors configured to generate sensor signals indicative of guest characteristics in an attraction measurement area of an individual attraction within an amusement park. The device also includes a controller configured to receive a user input to activate one or more light detection and ranging sensors to emit photons into an area; receive, at the light detection and ranging sensors, reflected photons; generate a sensor signal based on receiving the reflected photons; identify one or more anatomical features of a guest based in part on the sensor signals; estimate a dimension for the identified one or more anatomical features; and generate a guest height calculation based on the estimated dimension.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0010] FIG. 1 is a schematic view of an attraction measurement area of an amusement park in accordance with present embodiments;

[0011] FIG. 2 is a flow diagram of a method using a guest measurement system in accordance with present embodiments;

[0012] FIG. 3 is a block diagram of a guest measurement system in accordance with present embodiments;

[0013] FIG. 4 is a schematic view of an attraction measurement area of the amusement park utilizing the guest measurement system in accordance with present embodiments; and

[0014] FIG. 5 is a schematic view of a plurality of subregions of various limbs of the guest that are utilized as part of height calculation logic of the guest measurement system, in accordance with present embodiments.

DETAILED DESCRIPTION

[0015] Presently disclosed embodiments facilitate attraction measurement techniques for measuring one or more guest characteristics that in turn may be used to determine eligibility of the guest to enter a height-restricted attraction. The measurement techniques permit guests to be measured while in a variety of positions, e.g., standing, seated, slouching, or leaning over. Further, the measurement techniques reduce user measurement error or confusion associated with attraction eligibility requirements.

[0016] FIG. 1 shows an example guest measurement system 12 that includes a controller 14 coupled to one or more sensors 16 that generate guest measurement data in a guest measurement area 20 of an attraction. The one or more sensors 16 (e.g., depth sensors) are disposed about the area 20. The one or more sensors 16 may be co-located with one another or disposed separately at

distributed locations in the measurement area **20**. The one or more sensors may include depth sensors, color sensors, proximity sensors, and the like.

[0017] In one embodiment, the depth sensors may include an infrared projector, an infrared camera, and a RGB camera in order to obtain a guest's measurements. The one or more sensors may be co-located with one another or disposed separately at particular locations in the measurement area. For example, the one or more sensors may be disposed on a ceiling of the measurement area **20** or on a nearby wall in order to capture the measurements of the guest and surrounding area. Additionally, a guest-services employee may use a hand-held device containing the one or more sensors **16** in order to obtain measurements by moving the hand-held device in front of the guest to obtain relevant data. In one embodiment, the hand-held device containing the one or more sensors **16** is hand-held mobile device or tablet. The sensor **16** may include a light detection and ranging (LiDAR) capable sensing system. The LiDAR-capable sensing system may be integral with the hand-held mobile device or tablet to permit the hand-held mobile device or tablet to activate measurement at the LiDAR-capable sensing system, receive the measurement data using the LiDAR-capable sensing system, and view the The system may include a vertical-cavity surface-emitting laser, or VCSEL that emits bursts of photons in an area including the guest and that receives a reflected optical signal at the device. Measurement of the photons received at a detector (e.g., single-photon avalanche diodes (SPADs) adjacent to the source can be used to derive the distance of the objects or surfaces from the camera lens. The resolution of the distances can be based on an arrangement of the projection grid of the VCSEL. The device may, in embodiments, include a VCSEL that emits a series of points in the infrared which are detected by a receiver sensor. The LiDAR-capable sensing system can be used to generate depth information for the sensed area, which may lead to more accurate distance determination.

[0018] The one or more sensors **16** may also obtain measurements of a known height marker **32** disposed along an axis of a plane of the guest measurement area or known height prop disposed in the measurement area **20** in order to be calibrated, as explained further below.

[0019] The guest measurement system **12** permits guests to move naturally in or within the area **20** and may include a marker **38** for the guest to be measured automatically and, in embodiments, without guest input or estimation of their own measurements (e.g., height). The one or more sensors **16** collect environmental data that includes data representative of one or more guests, illustrated as a first guest **22a** and a second guest **22b** in FIG. **1**. The environmental data is provided to the controller **14**, which uses the environmental data to determine a guest characteristic, such as height. The guest measurement system **12** permits guests to be accurately measured while leaning over or seated, e.g., in a wheelchair or a stroller. As shown by way of example, the first guest **22a** is taller than the second guest **22b**. However, the first guest **22a** is leaning over, while the second guest **22b** is holding a balloon. If a distance **30** from the ground represents the minimum height to ride an attraction, an optical beam type sensor that measures beam disruptions at the minimum height would mistakenly indicate the second guest **22b** as being tall enough, because the balloon would break the beam. Further, the first guest **22a** would also be incorrectly identified as being not tall enough. In contrast, the disclosed measurement system **12** as provided herein would correctly identify the first guest **22a** as being tall enough and the second guest **22b** as not being at least the minimum height. The system **12** uses sensor signals to identify anatomical features and generate a height calculation based on estimated dimensions of the anatomical features. In embodiments, the system **12** may provide a display **36** or other indicator to indicate qualifying or not qualifying height calculations.

[0020] In an embodiment, using the information obtained via the one or more sensors **16**, one or more guest features are identified. The guest features may be an anatomical feature, e.g., each of the guest's limbs are identified. The guest's head, neck, torso, pelvis, arms, knees, shins, ankles, feet, hands, toes, fingers, and so forth may be identified. It may be appreciated that some of these limbs may have a paired limb or corresponding limb across a medial axis of the guest's body (i.e., a

right limb and a left limb). The presently disclosed embodiments include identifying a plurality of subregions for each of the identified anatomical features (e.g., limbs) and estimating a measurement or dimension of each of the plurality of subregions. The total height of the guest may be estimated or calculated by summing certain subregion measurements to obtain a limb estimate. For example, an estimated total leg height may be based on summing certain leg subregions. In an embodiment, the leg height plus the guest torso height and head dimensions may be used to determine guest height. In another embodiment, a guest's wing span can be used to determine that a guest's height has been calculated correctly. For example, the wing span may be generally defined as the length from one end of an individual's arms to the other when raised parallel to the ground at shoulder height at 90 degrees. The height of a human is generally about that of the wing span, which can serve as a quality check to confirm the estimated height of the guest.

[0021] Various mathematical relationships and equations (e.g., 3D Pythagorean theorem, linear interpolation) may be used to estimate the height of a particular subregion. It may be appreciated that a measurement of a particular subregion of a first limb may be substituted for a second corresponding limb, when a confidence level associated with the first limb exceeds that of the second corresponding limb by more than a threshold (e.g., more than one standard deviation). The presently disclosed embodiments include generating an overall height calculation of the guest based on the estimated limb measurements. In this way, the guest may remain seated (e.g., in a wheelchair) while being measured prior to accessing the ride.

[0022] With the foregoing in mind, FIG. 2 is a flow diagram of a method **100** using a guest measurement system in accordance with present techniques. The method **100** includes receiving (block **102**) sensor signals indicative of guest characteristics, e.g., depth sensor information, from one or more sensors disposed in a measurement area before boarding a ride.

[0023] The method **100** includes detecting (block **104**) anatomical features, e.g., one or more limbs, of a guest based on the information obtained from the one or more sensors. It may be appreciated that the sensor information obtained from the one or more sensors may first be filtered, normalized, calibrated, and/or pre-processed to be used more readily with a height calculation logic as provided herein. A presence and a position of various limbs and/or anatomical features of the guest, including the guest's head, neck, torso, pelvis, thigh, knee, calf, shin, ankle, foot, toe, hand, wrist, finger, elbow, forearm, and so forth may be identified or detected based on the sensor information. Skeletal tracking may be used to predict or observe the position of anatomical joints within a guest's body. The positions will be used to create a data structure from which the distances between joints can be calculated and visualized on a screen. Adding the lengths between the identified joints can be used to estimate height. In an embodiment, the techniques may be used to identify a limb difference, such as an amputation and the length of identified amputation. Based on stored attraction information, a notification can be generated for attractions that the guest is eligible to ride. The system may use depth sensors to create an image of skeletal structure of an individual within a 3D space in real-life coordinates. Multiple depth sensors can be used to create a composite image in 3D space that can reinforce the data collected.

[0024] The method **100** includes identifying a plurality of subregions for an individual limb (block **106**). For example, the method **100** may use various mathematical equations and relationships (e.g., geometry, a 3D Pythagorean theorem, linear interpolation) in order to estimate one or more dimensions of a particular subregion. It may be appreciated that the limbs may be divided into subregions for purposes of estimating their dimensions. For example, a left calf of the guest may be measured in three subregions, while corresponding measurements of the corresponding three subregions on a right calf of the guest may be measured. By doing so, the guest measurement system may be able to better estimate the overall height of the guest by using the most accurate measurements for a particular subregion, whether the measurement used is from the left calf of the guest or the right calf of the guest, as explained further below with reference to FIG. 5. The estimated dimensions may include a length estimate between joints (e.g., wrist-to-elbow; elbow-to-

shoulder, ankle-to-knee, knee-to-hip) and/or a width estimate in a plane parallel to a length axis. [0025] That is, an overall limb measurement or dimensions may be estimated (block **108**) by using some measurements from a left limb and some measurements from a right limb. An accuracy of the estimations for each of the limbs may also be determined. In order to determine a certainty of the estimated dimensions of each subregion of each limb, the guest measurement system may perform a calibration to compare to a measurement obtained by the one or more sensors to those with a calibration standard, such as the height of the known height marker or the known height prop. A statistical analysis may be performed to identify outliers in the estimates. In some embodiments, the measurements falling outside of a threshold (e.g., one standard deviation, two standard deviations) may be removed from the subregion measurements in order to improve an accuracy of the overall height calculation.

[0026] The method **100** includes generating (block **110**) an overall guest height calculation for the guest based on combinations of the estimated limb subregions. When there is more than one measurement for a particular subregion (e.g., a subregion of a left thigh vs. a subregion of a right thigh), the guest measurement system may then determine which measurement for the particular subregion has a higher degree of certainty associated with it and use the more certain measurement. Once each of the subregions of the limbs are estimated, the method includes generating the final height calculation. By using the method described herein, the guest may be measured to determine whether he can enter the ride without having to stand up from a wheelchair, assisted mobility device, stroller, and so forth. Further, it may be appreciated that the techniques described herein may also be applied to a guest that does not have any movement restrictions.

[0027] FIG. **3** is a block diagram of a guest measurement system **200** in accordance with present embodiments. The guest measurement system **200** includes a controller **202** that may be implemented in a computing device, a server, a distributed processor, or a cloud computing environment. The controller **202** includes a processor **204** and a memory **206**. The processor **204** may include one or more processing devices (processing or computing circuitry), and the memory **206** may include one or more tangible, non-transitory, machine-readable media. By way of example, such machine-readable media can include RAM, ROM, EPROM, EEPROM, or optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by the processor **204** or by other processor-based devices (e.g., mobile devices). The controller **202** may utilize one or more types of models **208** (e.g., software-based models executable by the processor **204**). For example, the models **208** may include geometry-based models, such as a 3-D Pythagorean Theorem model. Association techniques may be used to find relationship between variables. In one embodiment, a skeletal tracker **216** may establish a presence and/or a position of various skeleton joints and/or anatomical features, such as a the guest's head, hands, torso, center of mass, pelvis/hips, thighs, knees, shins, ankles, feet, and so forth.

[0028] The model **208** may receive various inputs **210** in order to perform estimations for each subregion of the guest's limbs. The inputs **210** may include ride information **212** and depth data **214**. The ride information **212** may include restrictions for a particular ride. For example, the ride may have particular height or weight limitations. Additionally, certain ride seats may only accommodate a certain depth for a passenger, and thus, a guest with certain disability equipment (e.g., a cane, a wheelchair, etc.) may not be able to go on the ride, may require the use of an additional seat (e.g., adjacent seats), or may require a particular seat on the ride that is suitable for the guest.

[0029] The depth data **214** may include data from one or more sensors. The one or more sensors may be disposed on a ceiling or a wall of an attraction measurement room or may be disposed in a hand-held device that is operated by a team member. The one or more sensors may include depth sensors, color sensors, proximity sensors, and the like. In one embodiment, the depth sensors may

include an infrared projector, an infrared camera, and a RGB camera in order to obtain a guest's measurements. The infrared projector, the infrared camera, and the RGB camera may be disposed in close proximity to each other (e.g., within a few centimeters of one another) or co-located with one another. As may be appreciated, the infrared projector may project infrared light patterns on nearby objects in the area, which are detected by the infrared camera. The infrared camera and/or the RGB camera capture the projected infrared light patterns. The cameras then send the video of the captured infrared light pattern to the depth sensor processor to determine the depth of each of the projected infrared light patterns (e.g., dots). In one embodiment, density of the projected infrared light pattern on a particular object can be correlated to distance from the projector. In an embodiment, the depth sensors may include a LiDAR sensing system that generates the depth data **214**.

[0030] Returning now to the discussion of the model **208**, the model **208** uses various mathematical equations and relationships (e.g., geometry, a 3D Pythagorean theorem, linear interpolation) in order to estimate a height or length of a particular subregion of a particular identified limb. The identified limbs may be divided into a number of subregions, based in part on a size of the limb, a certainty of the particular identified limb, and so forth. Certain limbs may be present on either side of a medial axis of the body that divides the guest's body into left and right sides. When an identified limb has a paired limb (i.e., a left shin on the left side of the body and a right shin on the right side of the body), the model **208** may estimate the height of both the left and the right limb, but only use the estimate for one limb if the left limb and the right limb are deemed to be sufficiently similar in size. In this instance, the model **208** may use the estimated height of only one limb (i.e., the left shin or the right shin) after determining which of the estimations is more accurate based on the depth data **214**. However, if the model **208** determines that the left and the right limb are outside of a particular threshold of similarity, the model **208** may use the estimate of the height of both of the left and the right limb. In another embodiment, the model **208** estimates and uses the height of both of the paired limbs, regardless of a threshold of similarity.

[0031] In one embodiment, the model **208** determines the height or the length of corresponding subregions for paired limbs. For example, the model **208** may determine the height or the length of a first region, a second region, and a third region of the guest's left calf. The model **208** may also determine the height or the length of a corresponding first region, a corresponding second region, and a corresponding third region of the guest's right calf. The model **208** then determines whether the first region of the left calf or the corresponding first region of the right calf is more accurate, based on a certainty of the depth data **214** provided from the region. The model **208** may then use the more accurate measurement to proceed with generating the overall height calculation. Indeed, the model **208** may use measurements from a particular subregion of either side of the body for one particular limb, in order to improve accuracy.

[0032] In order to determine a certainty of the estimated height of each subregion of each limb, the controller **202** may perform a calibration to compare to a measurement obtained by the one or more sensors to those with a calibration standard, such as the height of the known height marker or the known height prop. The controller **202** may then determine whether the certainty of the measured height of each subregion or limb is acceptable (e.g., within a determined acceptable error). When there is more than one measurement for a particular subregion (e.g., a subregion of a left thigh vs. a subregion of a right thigh), the controller **202** may then determine which measurement for the particular subregion has a higher degree of certainty associated with it and use the more certain measurement.

[0033] The model **208** may then use the more accurate measurement for the particular region in order to output a validated height calculation **218** for the particular region. Once the controller **202** has determined the validated height calculation **218** for each of the particular regions, an overall validated height calculation may be generated. The controller **202** may output the overall validated height calculation of the guest's entire body by summing each of the validated height estimations of

each limb. The final validated height calculation may be used to determine whether the ride has sufficient space to accommodate the guest while the guest remains seated or assisted by the guest's assistive equipment (e.g., in the guest's wheelchair, assisted mobility device, stroller) without requiring the guest to disembark from the guest's assistive equipment. In an embodiment, the output may include a notification regarding eligibility to ride a particular attraction based on the estimated height.

[0034] FIG. 4 is a schematic of a side view of an attraction measurement area of the theme park utilizing the guest measurement system **200** of FIG. 3 in accordance with present embodiments. In the illustrated embodiment, the guest **22** is able to be measured in the attraction measurement area **220** that is a separate enclosed room that is adjacent to the ride area **222**. The separate enclosed room may include a ceiling **224**, a floor **226**, and a plurality of walls **228** to enclose the measurement area. The attraction measurement area may include one or more sensors **232** within the attraction measurement area **220**. The one or more sensors **232** can be moved via a motor **234** that is communicatively coupled to a controller **202** in order to adequately capture the image data. Additionally or alternatively, the one or more sensors **232** can be disposed in a hand-held device **236** (e.g., a wand, a tablet, or a mobile device). The hand-held device **236** may be used by a park employee **238** to capture the image data and/or the depth data **214** when the hand-held device **236** is positioned in front of the guest **22**.

[0035] It may be appreciated that the guest **22** may be asked to move into a space **240** so that the guest **22** may be measured. The space **240** may be a distance **242** away from a height marker **244**. The space **240** may be an area where the space **240** can be readily viewed via the one or more sensors **232** that are disposed above the space **240** (e.g., along the rail **230** disposed along the ceiling **224** or via the hand-held device **236** held by the park employee). The height marker **244** may be a particular marker or a prop with a known height that is located the distance **242** behind the space **240**. The height marker **244** may be disposed substantially parallel to an axis **246**, where the axis **246** may be defined as the axis extending between the ceiling **224** and the floor **226**. When the guest **22** is determined to be in position, the height measurement can be initiated. In certain embodiments, the positioning of the guest within a threshold distance to the height marker **244** permits the sensor data to be calibrated to known measurements or dimensions of the height marker **244**. Once calibrated, the sensor data is then used to estimate dimensions of the anatomical features.

[0036] In embodiments, the sensors **232** are housed in a mobile device or tablet, and an operator activates the sensors **232** via interaction with a user interface of the mobile device or tablet.

[0037] It may be appreciated that the guest measurement system **200** may be used to measure the guests **22** that are in a wheelchair, an assisted mobility device, a stroller, a cart, and so forth, or any time the guest **22** is holding an accessory **248** (e.g., a cane, a shopping bag, a purse) that may partially obstruct the view of the guest **22**. In the illustrated embodiment, the guest measurement system **200** may be used to measure the guest **22** in a wheelchair. When the guest **22** is in a wheelchair or a stroller, it may be appreciated that a portion of the guest's body may be obstructed from view of the one or more sensors **232**. For example, a view of the guest's left leg may be obstructed by a component of the wheelchair. In this instance, the one or more sensors **232** may be unable to adequately capture the dimensions of the guest's left leg. As explained in further detail below, the guest measurement system **200** may then take one or more corrective actions, including but not limited to, attempting to re-measure the left leg, estimating the measurements of the left leg based on other body part measurements, substituting the measurement of the right leg, or a combination thereof. It may be appreciated that the guest measurement system **200** may refer to a database of previously acquired guest measurements to estimate the measurements of the guest. Similar corrective actions may be taken in an instance where a view of the guest's body is obstructed by a stroller, a shopping cart, a shopping bag, and so forth.

[0038] The controller **202** may be communicatively coupled to a display **250**, the one or more

sensors **232**, the height marker **244** and/or a nearby ride **252**. As discussed above, the controller **202** may receive various inputs **210** in order to adequately estimate the height of each subregion of the guest's limbs. The inputs **210** may include ride information **212**, depth data **214**, and the like of FIG. **3**. The ride information **212** may include a number of available seats, a number of seats that can accommodate guests in an assisted mobility vehicle, restrictions about which types of assisted mobility vehicles are suitable for the particular ride, height restrictions, and weight restrictions, among others. The depth data **214** may be received from the one or more sensors **232** disposed throughout the attraction measurement area. The depth data **214** may be obtained from infrared light patterns that are projected onto objects (e.g., the guest) by one or more infrared light projectors. The one or more infrared cameras may then detect the projected light patterns to determine a depth associated with the object that the pattern is detected on. Both of the various inputs **210** are then input into the model **208** in order to estimate the height of a particular subregion of an identified limb, as discussed above. The estimated height for the guest may be output to the display **250** in order to let the park employee know whether the guest **22** can be accommodated on the ride **252**.

[0039] In certain embodiments, the disclosed guest measurement system **200** may use facial recognition or other guest tracking techniques to correlate a guest to a measured height. Thus, the guest may be measured at an earlier point in the day, and the estimated height logged to the guest profile. Using guest recognition techniques, the recognized guest can be associated with their estimated height using the guest profile at other attractions to determine ride eligibility. In this manner, the measurement need only occur once during the day.

[0040] FIG. **5** is a schematic view of a plurality of subregions **260** of various limbs of the guest **22** that are utilized in the guest measurement system, in accordance with present embodiments. It may be appreciated that each of the identified limbs may be split into various subregions for purposes of gathering the depth information. In some embodiments, the paired limbs (e.g., a left shin and a right shin) may be split into a varying number of subregions depending on a size of the limb. For example, the limb of a larger guest (e.g., having a longer shin) may be split into four subregions, whereas the limb of an average sized guest (e.g., having an average length shin **261**) may be split into only three subregions. Though certain anatomical features will not have pairs (e.g., a torso, a neck), the same concept of applying a varying number of subregions to the identified limb may apply. In the illustrated embodiment, though the torso **262** of the guest **22** is split into two subregions, the depth information for the torso may be obtained over three, four, or more subregions for a larger guest.

[0041] As discussed above, the guest measurement system **200** of FIG. **3** may receive depth information from the one or more sensors **232** of FIG. **4**. The guest measurement system **200** may receive multiple depth information readings for a given subregion from various sensors disposed in the attraction measurement area **220** of FIG. **4**. For example, for the particular subregion **260** (i.e., the torso **262**), the guest measurement system **200** may receive depth information from the one or more sensors **232** located above the space **240** of FIG. **4**, on the adjacent walls **224**, **228** of FIG. **4**, and/or from the hand-held device **236** of FIG. **4**. It may be appreciated that though the guest measurement system **200** receives depth information for the same particular subregion from the various sensors **232**, the depth information from each sensor **232** may remain siloed. In this way, if the depth information received from a single sensor **232** appears to be erroneous (e.g., outside an acceptable measure of uncertainty), the guest measurement system **200** may discount the erroneous depth information when estimating the length of the particular limb.

[0042] In some embodiments, the guest measurement system **200** may receive multiple depth information measurements for each subregion **260**. The guest measurement system **200** may then determine whether any of the collected depth information includes one or more outliers for the particular subregion. When the guest measurement system **200** determines a presence of any statistical outliers, the outliers may be excluded from the collected depth information used to

estimate the height of the particular subregion **260**. When the guest measurement system **200** determines that one or more outliers are present, one or more corrective actions may be implemented. The one or more corrective actions may be implemented in place of or in addition to excluding the outlier. As discussed above, the one or more corrective actions may include, but are not limited to, attempting to re-measure the particular subregion **260** of the identified limb, estimating the measurements of the particular subregion **260** of the identified limb based on other body part measurements (e.g., estimating the measurement of the particular subregion **260** of the identified limb based on a database), or substituting the measurement of the paired subregion **260** of the identified limb.

[0043] For example, it may be appreciated that the guest measurement system **200** may re-calibrate the one or more sensors **232** in the event that one or more outliers is identified. Re-calibration of the one or more sensors **232** may include comparing a measurement obtained by the one or more sensors **232** to those with a known standard, such as the height of the known height marker **244**, or another suitable calibration technique. In another non-limiting example, the corrective action may include estimating the measurement of the particular subregion of the identified limb based on measurements obtained from a historical database. As such, the guest measurement system **200** may substitute one or more measurements from the historical database by identifying suitable subregion measurements based in part on the guest's overall height and stature, whether the guest is sitting or standing, a particular model of an assisted mobility vehicle (e.g., wheelchair, stroller), and the like.

[0044] In another non-limiting example, the guest measurement system **200** may use subregions having higher quality measurements and may discard or not use subregions having lower quality measurements. Thus, the measurement or estimate of the guest limb length may be based on a summation of individual subregions from different corresponding limbs. In one example, a leg measurement may be based on a summation of a left shin and a right thigh region if those subregions represent the highest quality subregions for the legs. The quality metric may be based on the confidence of the raw sensor data. For example, the guest measurement system **200** may substitute sensor data for a subregion of a left shin with a corresponding subregion of a right shin, when the sensor data for the right shin is identified as an outlier outside of the desired confidence interval. In this case, the guest measurement system **200** may disregard the data for the subregion of the left shin when generating the overall height calculation for the guest **22**.

[0045] While only certain features of the present embodiments have been illustrated and described herein, many modifications and changes will occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the present disclosure. Further, it should be understood that certain elements of the disclosed embodiments may be combined or exchanged with one another.

[0046] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A guest measurement system, comprising: one or more sensors configured to generate sensor signals indicative of a guest distance from a height marker in an attraction measurement area; and a controller configured to receive one or more sensor signals indicative of the guest distance from the one or more sensors, the controller comprising a processor configured to: determine that a guest is

within a threshold distance from the height marker based on the one or more sensor signals; and in response to determining that the guest is within a threshold distance from the height marker, initiate a guest measurement, wherein the guest measurement comprises using the one or more sensor signals to identify limbs of the guest, estimating dimensions of subregions of the limbs of the guest based on the height marker, and using the dimensions of the subregions to generate a guest height calculation.

2. The guest measurement system of claim 1, wherein the one or more sensors comprises a camera or a depth sensor.
3. The guest measurement system of claim 1, wherein the one or more sensors identify the limbs of the guest by determining a right side limb and a corresponding left side limb of a same type.
4. The guest measurement system of claim 3, wherein the processor is configured to: determine a first confidence interval of first dimensions for the right side limb and a second confidence interval of second dimensions for the corresponding left side limb; and determine that the first confidence interval or the second confidence interval exceeds a threshold.
5. The guest measurement system of claim 4, wherein the processor is configured to: upon determining that the first confidence interval exceeds the threshold and the second confidence interval does not exceed the threshold, use the first dimensions for subregions of the right side limb and not the second dimensions for subregions of the corresponding left side limb for determining the guest height calculation.
6. The guest measurement system of claim 1, wherein the processor is configured to: determine a length dimension of a length between joints of the limbs of the guest.
7. The guest measurement system of claim 1, wherein the limbs of the guest comprise a leg or an arm.
8. The guest measurement system of claim 1, wherein the processor is configured to: generate an indication that the guest is qualified for an attraction of an amusement park based in part on the guest height calculation.
9. The guest measurement system of claim 1, wherein the processor is configured to: determine that the guest distance exceeds the threshold distance from the height marker; and output an indication that the guest is not located within the threshold distance.
10. The guest measurement system of claim 1, wherein the processor is configured to: determine assisted ride eligibility for the guest, wherein assisted ride eligibility comprises the guest remaining seated or assisted by an assistive equipment, wherein the assistive equipment comprises a wheelchair, an assisted mobility device, a stroller, or any combination thereof, without requiring the guest to disembark from the assistive equipment based in part on the guest height calculation.
11. A method of measuring a guest in an attraction measurement area comprising: receiving one or more sensor signals indicative of a guest distance from a height marker from one or more sensors; identifying one or more anatomical features of a guest based at least in part on the sensor signals; estimating a dimension for the one or more anatomical features; generating a guest height calculation based on the dimension; and storing the guest height calculation as part of a guest profile of the guest, wherein the guest height calculation occurs once over a period of time.
12. The method of claim 11, comprising: identifying the guest based on guest tracking information; accessing the guest profile of the guest; and generating indications of ride eligibility at a plurality of rides using the stored guest height calculation of the guest profile.
13. The method of claim 11, wherein the height marker is disposed of along an axis of a plane of the attraction measurement area, and wherein the height marker comprises an object of a known height.
14. The method of claim 11, comprising: identifying a plurality of subregions for an individual anatomical feature of the one or more anatomical features; summing a portion of the plurality of subregions to generate a limb measurement; and generating the guest height calculation based on the limb measurement.

- 15.** The method of claim 11, wherein the one or more sensors comprises a camera.
- 16.** The method of claim 11, comprising: selecting a ride seat for the guest based in part on the guest height calculation.
- 17.** A guest measurement device, comprising: one or more sensors configured to generate sensor signals indicative of guest characteristics in an attraction measurement area; and a processor configured to: activate one or more light emitters to emit photons into an area; receive, at the one or more sensors, reflected photons; generate a sensor signal based on receiving the reflected photons; identify one or more anatomical features of a guest based in part on the sensor signals; estimate a dimension for the one or more anatomical features; and generate a guest height calculation based on the dimension.
- 18.** The guest measurement device of claim 17, wherein the processor is configured to: identify a plurality of subregions for an individual anatomical feature of the one or more anatomical features.
- 19.** The guest measurement device of claim 18, wherein the processor is configured to: estimate dimensions for the plurality of subregions of the individual anatomical feature.
- 20.** The guest measurement device of claim 17, wherein the processor is configured to: identify the one or more anatomical features of the guest by determining a right side limb and a corresponding left side limb of a same type.
-